

AC Series

Energy Storage Converter

Modbus RTU communication protocol

Revised record

Time	Version	Description	Monitor Version	Author
2022-05-31	V0.0.1	First draft		honghq
2022-08-12	V0.0.2	Adjustment of parameter settings		honghq
2022-12-21	V0.0.3	MAC classification and off-grid mode and control mode, alarm and fault information splitting, deleting useless alarms, adding parallel communication faults, setting parameters to increase the photovoltaic power low shutdown and charging mode, open-loop enable on the remote control, the rated power, phase voltage range is adjusted to a larger size, and part of the parameter default value adjustment.		honghq
2023-04-05	V0.0.4	Add authorized capacity, VSG parameters, standby control.		honghq
2023-07-22	V0.0.5	Grid-connected and off-grid mode is split into grid-connected and off-grid command, grid-connected and off-grid switching, control mode is divided into grid-connected and off-grid control mode, rated power and authorized capacity range is relaxed, active power, bus voltage reference, bus over-voltage point is adjusted to a smaller range.		honghq
2023-11-24	V0.0.6	The setting range of active and reactive power is liberalized to -1200~1200.		honghq
2024-8-19	V0.0.7	Updated the way of indicating version number; increased the setting range of DC bus over-voltage point and raised the default to 950V; added total discharge and total charge; relaxed the address range to 0~14; and modified the baud rate to a fixed value of 9600.		Pengxiangyu
2024-8-20	V1.0	Optimize the document format and relax the limits of parameter battery voltage and current.		honghq
2024-9-18	V1.0.1	Add the setting of module address in the control parameter under parameter setting, add the address of communication parameter setting and add dry contact control in it.		Pengxiangyu
2024-9-23	V1.0.2	Added EMS communication failure, BMS communication failure, BMS failure, dry contact 1 failure, dry contact 2 failure in alarm message 7. Added BMSenable bit and BMS communication failure shutdown enable bit to communication parameter settings.		Pengxiangyu

2024-12-17	V1.0.3	Set BMS enable bit default value to disable		Pengxiangyu
2024-04-03	V1.0.4	Add communication failure confirmation time		Pengxiangyu
2025-04-22	V1.0.5	Added communication fault confirmation time		Pengxiangyu
2024-12-17	V2.0.0	Added SN-related points		honghq
2025-05-28	V2.1.0	Improved status information; changed battery float charging voltage and equalization charging voltage to battery charging limit voltage and discharge limit voltage; adjusted primary frequency and voltage regulation parameter ranges; changed charging method to power imbalance control; added parameters for cabinet address, battery SOC, and battery SOC sag		honghq

Contents

1. Introductory	1
2. Physical interface	1
2.1. Communication interface	1
2.2. Information transmission	1
2.3. Data transmission rate	1
2.4. Frame interval time requirement	1
2.5. Communication mode	1
2.6. Modbus node number	1
2.7. CRC Check	1
3. Physical interface	3
3.1. FC3/FC4 Block Read	3
3.2. FC5 single coil write	3
3.3. FC6 word write	4
3.4. FC16 block write	5
3.5. Error return frame	5
4. Protocol text	7
4.1. Basic Product Information - Function 03 or 04	7
4.2. Analog Information (Telemetry) - Function 03 or 04	7
4.3. Alarm volume (telemetry) - Function 03 or 04	10
4.4. Status Quantity (Telecommand) - Function 03 or 04	14
4.5. Parameter setting (remote control) - Function 03 or 04 or 05 or 06 or 16	15

4.6. Set date and time (remote control) — Function 03 or 16.....23

4.7. Remote On/Off (Remote Control) - Function 05 or 06..... 24

1. Introductory

Designed to realize the serial communication between the energy storage converter and the host of communication equipment such as EMS, this protocol allows the converter to be connected to the MODBUS RTU system.

2. Physical interface

2.1. Communication interface

RS485 is adopted.

2.2. Information transmission

Asynchronous mode, start bit 1 bit, data bit 8 bits, stop bit 1 bit, no parity bit.

2.3. Data transmission rate

Baud rate 9600bps.

2.4. Frame interval time requirement

Communication frame interval of 100ms or more is required.

2.5. Communication mode

Adopting master-slave communication mode, the storage converter plays the role of communication slave and waits for the Modbus host to query.

2.6. Modbus node number

The physical address of the converter of IP20 is set by dialing code, the setting range is 0~14, corresponding to MODBUS node number 1~15, the factory default dialing code is set to 0, corresponding to MODBUS node number is 1 by default ; the module address range of the converter of IP65 is 1~15, corresponding to MODBUS node number 1~15, the factory default is 1, and it is set by telecontrol, please refer to 4.5.1 for details of the setting. When communicating to a single IP65 module, use Modbus node number 160, ID number 3182.

2.7. CRC Check

This protocol adopts the standard calculation method of Modbus RTU CRC16,

and the checking order is (Low-High), low byte first, then high byte.

- The CRC code is calculated by:

(1) Preset a 16-bit register as hexadecimal FFFF (i.e., all 1s); call this register the CRC register;

(2) Isolate the first 8-bit binary data (the first byte of the communication information frame) with the lower 8 bits of the 16-bit CRC register, and put the result in the CRC register;

(3) Shift the contents of the CRC register one bit to the right (toward the lower bit) to fill the highest bit with a 0, and check the shifted out bit after the right shift;

(4) If the shifted out bit is 0: repeat step 3 (shift right one bit again); if the shifted out bit is 1: the CRC register is iso-orthogonal to the polynomial A001 (1010 0000 0000 0001);

(5) Repeat steps 3 and 4 until it is shifted right 8 times, so that the entire 8-bit data is processed in its entirety;

(6) repeating steps 2 through 5 for the next byte of the communication information frame;

(7) After all bytes of the communication information frame are calculated according to the above steps, the high and low bytes of the 16-bit CRC register are exchanged;

(8) The content of the CRC register obtained at last is: CRC code.

3. Physical interface

3.1. FC3/FC4 Block Read

Query Frame:

NODE	FUN	ID(start)	NR	CRC
<i>1 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2bin</i>	<i>2 bin</i>

NODE MODBUS Node Number

FUN Function Code, FC3=3 or FC4=4

ID(start) The first address of the ID of the queried data area must be the ID defined in the protocol text

NR Number of IDs in the Data Area Being Queried

CRC Checksum, L-H Mode

Feedback Frame:

NODE	FUN	NR_BYTE	NR	CRC
<i>1 bin</i>	<i>1 bin</i>	<i>1 bin</i>	<i>n bin</i>	<i>2 bin</i>

NODE MODBUS Node Number

FUN Function Code, FC3=3 or FC4=4

NR_BYTE Number of bytes of feedback data DATA, equal to twice NR

DATA Feedback data area, length equal to NR_BYTE

CRC Checksum, L-H Mode

3.2. FC5 single coil write

Query Frame:

NODE	FUN	ID	DATA	CRC
<i>1 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2bin</i>	<i>2 bin</i>

NODE MODBUS node number

FUN Function Code, FC5=5

ID The ID address of the written data must be the ID defined in the protocol text.

DATA The data to be written, 0xFF00 means ON, 0x0000 means OFF

CRC Checksum, L-H method allowed

Query Frame:(return as is)

NODE	FUN	ID	DATA	CRC
<i>1 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2bin</i>	<i>2 bin</i>

NODE MODBUS node number

FUN Function Code, FC5=5

ID The ID address of the written data must be the ID defined in the protocol text.

DATA Data to be written

CRC Checksum, L-H method allowed

3.3. FC6 word write

Query Frame:

NODE	FUN	ID	DATA	CRC
<i>1 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2bin</i>	<i>2 bin</i>

NODE MODBUS node number

FUN Function Code, FC6=6

ID The ID address of the written data must be the ID defined in the protocol text.

DATA Data to be written

CRC Checksum, L-H method allowed

Feedback Frame:(return as is)

NODE	FUN	ID	DATA	CRC
<i>1 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2bin</i>	<i>2 bin</i>

NODE MODBUS node number

FUN Function Code, FC6=6

ID The ID address of the written data must be the ID defined in the protocol text.

DATA Data to be written

CRC Checksum, L-H method allowed

3.4. FC16 block write

Query Frame:

NODE	FUN	ID(start)	NR	NR_BYTE	DATA	CRC
<i>1 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2bin</i>	<i>1 bin</i>	<i>n bin</i>	<i>2 bin</i>

NODE	MODBUS node number
FUN	Function Code, FC16=16
ID(start)	The first address of the ID written to the data area must be the ID defined in the protocol text
NR	Number of IDs written to the data area, legal range 1 to 123, and no undefined IDs in the coverage area
NR_BYTE	The length of DATA written to the data area, must be twice as long as NR
DATA	The length of the DATA area written to, equal to NR_BYTE
CRC	Checksum, L-H method allowed

Feedback Frame:

NODE	FUN	ID(start)	NR	CRC
<i>1 bin</i>	<i>1 bin</i>	<i>2 bin</i>	<i>2bin</i>	<i>2 bin</i>

NODE	MODBUS node number
FUN	Function Code, FC16=16
ID(start)	The first address of the ID written to the data area must be the ID defined in the protocol text
NR	The number of IDs written to the data area, the legal range is 1 to 123, and the coverage cannot have undefined IDs
CRC	Checksum, L-H method allowed

3.5. Error return frame

NODE	FUN	ERROR	CRC
<i>1 bin</i>	<i>1 bin</i>	<i>1 bin</i>	<i>2 bin</i>

NODE	MODBUS node number
FUN	Function number, the highest position of the function number, 1, indicates an exception
ERROR	Error code Error code 01 = Unknown function code, the received function

code is not supported.

Error code 02 = Address error, the received address (ID) is out of range.

Error code 03 = Data error, received data format not recognized, CRC checksum error.

Error code 06 = device busy, data not ready

CRC Checksum, L-H method allowed

4. Protocol text

4.1. Basic Product Information - Function 03 or 04

<i>ID</i>	<i>Byte</i>	<i>Item</i>	<i>Value Type</i>	<i>Format</i>
1000	2	ARM major version number	UINT16	HEX
1001	2	ARM minor version number	UINT16	HEX
1002	2	ARM Version Revision Number	UINT16	HEX
1003	2	ARM version protocol number	UINT16	HEX
1004	2	DSP major version number	UINT16	HEX
1005	2	DSP minor version number	UINT16	HEX
1006	2	DSP Version Revision Number	UINT16	HEX
1007	2	CPLD software version number	UINT16	HEX

4.2. Analog Information (Telemetry) - Function 03 or 04

4.2.1. System analog

<i>ID</i>	<i>Byte</i>	<i>Item</i>	<i>Value</i>	<i>Value Type</i>	<i>Format</i>
1050	2	Grid voltage A(AB)	2200corresponds to220.0V	UINT16	HEX
1051	2	Grid voltage B(BC)	2200corresponds to220.0V	UINT16	HEX
1052	2	Grid voltage C(CA)	2200corresponds to220.0V	UINT16	HEX
1053	2	Grid current A	1000corresponds to100.0A	UINT16	HEX
1054	2	Grid current B	1000corresponds to100.0A	UINT16	HEX
1055	2	Grid current C	1000corresponds to100.0A	UINT16	HEX
1056	2	Grid frequency A	5000corresponds to50.00Hz	UINT16	HEX
1057	2	Grid frequency B	5000corresponds to50.00Hz	UINT16	HEX
1058	2	Grid frequency C	5000corresponds to50.00Hz	UINT16	HEX
1059	2	Total power factor	100corresponds to1.00	INT16	HEX
1060	2	Total active power	1000corresponds to100.0kW	INT16	HEX
1061	2	Total reactive power	1000corresponds to100.0kvar	INT16	HEX
1062	2	Total apparent power	1000corresponds to100.0kVA	UINT16	HEX

1063	2	DC voltage	1000corresponds to100.0V	UINT16	HEX
1064	2	DC current	1000corresponds to100.0A	INT16	HEX
1065	2	DC power	1000corresponds to100.0kW	INT16	HEX
1066	2	Busbar voltage	1000corresponds to100.0V	UINT16	HEX
1067	2	Positive bus voltage	1000corresponds to100.0V	UINT16	HEX
1068	2	Negative bus voltage	1000corresponds to100.0V	UINT16	HEX
1069	2	Balanced bridge current	1000corresponds to100.0A	INT16	HEX
1070	2	Ground-to-negative voltage	1000corresponds to100.0V	INT16	HEX
1071	2	Positive to ground impedance	100corresponds to100k Ω	UINT16	HEX
1072	2	Negative Impedance to Ground	100corresponds to100k Ω	UINT16	HEX
1073	2	Leakage Current	100corresponds to1.00A	UINT16	HEX
1074	2	Power Tube Temperature	100corresponds to100°C	INT16	HEX
1075	2	Balance Bridge Temperature	100corresponds to100°C	INT16	HEX
1076	2	Ambient temperature	100corresponds to100°C	INT16	HEX
1077	2	Converter efficiency	1000corresponds to100.0%	UINT16	HEX
1078	2	Power Factor A	100corresponds to1.00	INT16	HEX
1079	2	Power Factor B	100corresponds to1.00	INT16	HEX
1080	2	Power factor C	100corresponds to1.00	INT16	HEX
1081	2	Active power A	1000corresponds to100.0kW	INT16	HEX
1082	2	Active power B	1000corresponds to100.0kW	INT16	HEX
1083	2	Active power C	1000corresponds to100.0kW	INT16	HEX
1084	2	Reactive power A	1000corresponds to100.0 kvar	INT16	HEX
1085	2	Reactive power B	1000corresponds to100.0 kvar	INT16	HEX
1086	2	Reactive power C	1000corresponds to100.0 kvar	INT16	HEX
1087	2	Apparent power A	1000corresponds to100.0kVA	UINT16	HEX
1088	2	Apparent power B	1000corresponds to100.0kVA	UINT16	HEX
1089	2	Apparent power C	1000corresponds to100.0kVA	UINT16	HEX

1090	2	Inverter voltage A(AB)	2200corresponds to220.0V	UINT16	HEX
1091	2	Inverter voltage B(BC)	2200corresponds to220.0V	UINT16	HEX
1092	2	Inverter voltage C(CA)	2200corresponds to220.0V	UINT16	HEX
1093	2	Inverter current A	1000corresponds to100.0A	UINT16	HEX
1094	2	Inverter current B	1000corresponds to100.0A	UINT16	HEX
1095	2	Inverter current C	1000corresponds to100.0A	UINT16	HEX
1096	2	Reserved	2200corresponds to220.0V	INT16	HEX
1097	2	Reserved	2200corresponds to220.0V	INT16	HEX
1098	2	Reserved	2200corresponds to220.0V	INT16	HEX
1099	2	Reserved	1000corresponds to100.0A	INT16	HEX
1100	2	Reserved	1000corresponds to100.0A	INT16	HEX
1101	2	Reserved	1000corresponds to100.0A	INT16	HEX
1102	2	Reserved	1corresponds to1	UINT16	HEX
1103	2	Reserved	1corresponds to1	UINT16	HEX
1104	2	Reserved	1corresponds to1	UINT16	HEX
1105 1106	4	Total charge (1)	1000corresponds to100.0kWh	UINT32	HEX Hi-Lo
1107 1108	4	Total discharge (1)	1000corresponds to100.0kWh	UINT32	HEX Hi-Lo

Notes:

(1) The total charging and discharging amount accumulated is the total power value from this power-up to the current moment, and the initial value is reset to 1 for the next power-up.

4.2.2. Telemetry Example

Example: Modbus node number 01, read three analog quantities starting from telemetry information ID 1050, data order H-L, checksum order L-H.

Telemetry (HEX): 01 03 04 1A 00 03 25 3C

Where: 01 - Modbus number 01; 03 - function code 03;
04 1A - starting ID of 1050; 00 03 - quantity 3;
25 3C - CRC16 checksum

Answer (HEX): 01 03 06 08 98 08 98 08 98 84 04

Where: 01 - Modbus number 01; 03 - function code 03;
 06——Data byte count 08 98——220V grid voltage A(AB)
 08 98——220V grid voltage B(BC)
 08 98——220V grid voltage C(CA)
 84 04——CRC16 checksum

4.3. Alarm volume (telemetry) - Function 03 or 04

4.3.1. Alarm message 1

<i>ID</i>	<i>Bit</i>	<i>Item</i>	<i>Value</i>
2050	<i>Bit 0</i>	<i>Positive bus over-pressure</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 1</i>	<i>Negative bus over-voltage</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 2</i>	<i>DC bus over-voltage</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 3</i>	<i>Bus half-voltage unbalance</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 4</i>	<i>DC bus short circuit</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 5</i>	<i>DC over-current</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 6</i>	<i>Balance bridge over-current</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 7</i>	<i>DC Voltage Reverse</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 8</i>	<i>Low DC Voltage</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 9</i>	<i>DC voltage high</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit10</i>	<i>Abnormal insulation impedance</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 11</i>	<i>PV power low shutdown</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 12</i>		
	<i>Bit 13</i>		
	<i>Bit 14</i>		
	<i>Bit 15</i>		

4.3.2. Alarm message 2

<i>ID</i>	<i>Bit</i>	<i>Item</i>	<i>Value</i>
2051	<i>Bit 0</i>	<i>Abnormal phase A inverter voltage</i>	<i>"0: Normal" "1:Malfunction"</i>

	Bit 1	Abnormal phase B inverter voltage	"0: Normal" "1:Malfunction"
	Bit 2	Abnormal phase C inverter voltage	"0: Normal" "1:Malfunction"
	Bit 3	A-Phase inverter voltage direct current abnormal	"0: Normal" "1:Malfunction"
	Bit 4	B-phase inverter voltage direct current abnormal	"0: Normal" "1:Malfunction"
	Bit 5	C-Phase inverter voltage direct current abnormal	"0: Normal" "1:Malfunction"
	Bit 6	A-Phase output overload shutdown	"0: Normal" "1:Malfunction"
	Bit 7	B-phase output overload shutdown	"0: Normal" "1:Malfunction"
	Bit 8	C-phase output overload shutdown	"0: Normal" "1:Malfunction"
	Bit 9	A-phase output over-current	"0: Normal" "1:Malfunction"
	Bit10	B-phase output over-current	"0: Normal" "1:Malfunction"
	Bit 11	C-Phase output over-current	"0: Normal" "1:Malfunction"
	Bit 12	A-phase output short circuit	"0: Normal" "1:Malfunction"
	Bit 13	B-phase output short circuit	"0: Normal" "1:Malfunction"
	Bit 14	C-phase output short circuit	"0: Normal" "1:Malfunction"
	Bit 15	Inverter phase out of synchronization	"0: Normal" "1:Malfunction"

4.3.3. Alarm message 3

ID	Bit	Item	Value
2052	Bit 0	Grid over-voltage	"0: Normal" "1:Malfunction"
	Bit 1	Grid under-voltage	"0: Normal" "1:Malfunction"
	Bit 2	Grid over-frequency	"0: Normal" "1:Malfunction"
	Bit 3	Grid under-frequency	"0: Normal" "1:Malfunction"
	Bit 4	Islanding protection	"0: Normal" "1:Malfunction"
	Bit 5	Grid wrong phase	"0: Normal" "1:Malfunction"
	Bit 6	AC Power Failure	"0: Normal" "1:Malfunction"
	Bit 7	AC current limiting shutdown	"0: Normal" "1:Malfunction"
	Bit 8	Parallel Cable Fault	"0: Normal" "1:Malfunction"
	Bit 9	Carrier Synchronization Fault	"0: Normal" "1:Malfunction"

	<i>Bit10</i>	<i>Inverter Synchronization Fault</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 11</i>	<i>Parallel Communication Fault</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 12</i>	<i>AC Fuse Failure</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 13</i>	<i>Power Tube Over Temperature</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 14</i>	<i>Power Supply Fault</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 15</i>	<i>Leakage current fault</i>	<i>"0: Normal" "1:Malfunction"</i>

4.3.4. Alarm message 4

<i>ID</i>	<i>Bit</i>	<i>Item</i>	<i>Value</i>
2053	<i>Bit 0</i>	<i>DC Precharge Fault</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 1</i>	<i>AC Precharge Fault</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 2</i>	<i>DC Relay Short Circuit</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 3</i>	<i>DC Relay Open Circuit</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 4</i>	<i>AC A-phase Relay Short Circuit</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 5</i>	<i>AC B-phase relay short circuit</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 6</i>	<i>AC C-phase relay short circuit</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 7</i>	<i>AC A-phase relay open</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 8</i>	<i>AC B-phase relay open</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 9</i>	<i>AC C-phase relay open</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit10</i>	<i>A-phase bridge arm straight through</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 11</i>	<i>B-phase bridge arm through</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 12</i>	<i>C-phase bridge arm through</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 13</i>		
	<i>Bit 14</i>		
	<i>Bit 15</i>		

4.3.5. Alarm message 5

<i>ID</i>	<i>Bit</i>	<i>Item</i>	<i>Value</i>
2054	<i>Bit 0</i>	<i>Grid Current Zero Bias Abnormal</i>	<i>"0: Normal" "1:Malfunction"</i>

	<i>Bit 1</i>	<i>Inverter current zero bias anomaly</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 2</i>	<i>Inverter current direct current zero bias anomaly</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 3</i>	<i>Zero bias DC current anomaly</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 4</i>	<i>Balance bridge current zero bias anomaly</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 5</i>	<i>Leakage current zero bias anomaly</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 6</i>	<i>2.5V reference abnormal</i>	<i>"0: Normal" "1:Malfunction"</i>
	<i>Bit 7</i>		
	<i>Bit 8</i>		
	<i>Bit 9</i>		
	<i>Bit10</i>		
	<i>Bit 11</i>		
	<i>Bit 12</i>		
	<i>Bit 13</i>		
	<i>Bit 14</i>		
	<i>Bit 15</i>		

4.3.6. Alarm message 6

<i>ID</i>	<i>Bit</i>	<i>Item</i>	<i>Value</i>
2055	<i>Bit 0</i>	<i>A-phase output overload alarm</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 1</i>	<i>B-phase output overload alarm</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 2</i>	<i>C-Phase output overload alarm</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 3</i>	<i>Low voltage ride-through</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 4</i>	<i>High voltage ride-through</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 5</i>	<i>Balanced bridge wave-by-wave current limit alarm</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 6</i>	<i>Balance bridge over temperature</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 7</i>	<i>High ambient temperature</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 8</i>	<i>Over Temperature Derating</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 9</i>	<i>DC lightning protection fault</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit10</i>	<i>AC lightning protection fault</i>	<i>"0: Normal" "1:Alarm"</i>

	<i>Bit 11</i>	<i>Fan fault 1</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 12</i>	<i>Fan fault 2</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 13</i>	<i>Battery over-charge</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 14</i>	<i>Battery over-discharge</i>	<i>"0: Normal" "1:Alarm"</i>
	<i>Bit 15</i>		

4.3.7. Alarm message 7

<i>ID</i>	<i>Bit</i>	<i>Item</i>	<i>Value</i>
2056	<i>Bit 0</i>	<i>Internal Communication Failure</i>	<i>"0: No" "1: Yes"</i>
	<i>Bit 1</i>	<i>EMS communication failure</i>	<i>"0: No" "1: Yes"</i>
	<i>Bit 2</i>	<i>BMS communication failure</i>	<i>"0: No" "1: Yes"</i>
	<i>Bit 3</i>	<i>BMS fault</i>	<i>"0: No" "1: Yes"</i>
	<i>Bit 4</i>	<i>Dry contact 1 fault</i>	<i>"0: No" "1: Yes"</i>
	<i>Bit 5</i>	<i>Dry contact 2 fault</i>	<i>"0: No" "1: Yes"</i>
	<i>Bit 6</i>		
	<i>Bit 7</i>		
	<i>Bit 8</i>		
	<i>Bit 9</i>		
	<i>Bit10</i>		
	<i>Bit 11</i>		
	<i>Bit 12</i>		
	<i>Bit 13</i>		
	<i>Bit 14</i>		
	<i>Bit 15</i>		

4.4. Status Quantity (Telecommand) - Function 03 or 04

4.4.1. Status message 1

<i>ID</i>	<i>Bit</i>	<i>Item</i>	<i>Value</i>
-----------	------------	-------------	--------------

2057	Bit 0	DC Pre-Charge	"0: Disconnected" "1: Connected"
	Bit 1	AC Soft Start	"0: Disconnected" "1: Connected"
	Bit 2	DC Relay	"0: Disconnected" "1: Connected"
	Bit 3	AC Relay	"0: Disconnected" "1: Connected"
	Bit 4	Off-grid	"0: On-Grid" "1: Off-Grid"
	Bit 5	Charge and Discharge	"0: Discharging" "1: Charging"
	Bit 6	Operation Status	"0: Stop" "1: Run"
	Bit 7	Fault status	"0: No" "1: Yes"
	Bit 8	Master-slave machine	"0: Slave machine" "1: Master machine"
	Bit 9	Master-slave cabinet	"0: Slave cabinet" "1: Master cabinet"
	Bit10	Standby	"0: No" "1: Standby"
	Bit 11	Shutdown	"0: No" "1: Shutdown"
	Bit 12	EPO	"0: No" "1: Emergency stop"
	Bit 13		
	Bit 14		
	Bit 15	Parameter initialization	"0: Not completed" "1: Completed"

4.4.2. Example of telecommunication

Example: Device node number (device number) 01, reading telecommand information with a start ID of 2050, quantity of 1, data order H-L, checksum order L-H.

Telecommand (HEX): 01 03 08 02 00 01 27 AA

Answer (HEX): 01 03 02 04 00 BA 84

4.5. Parameter setting (remote control) - Function 03 or 04 or 05 or 06 or 16

4.5.1. Control parameter setting

ID	Byte	Item	Value	Value Type	Format
3050	2	Total active power	-1200~1200 (-120%~120%) (default 50)	INT16	HEX
3051	2	Total reactive power	-1200~1200 (-120%~120%) (default 0)	INT16	HEX
3052	2	Total power factor	-1000~1000 (-1~1)	INT16	HEX

			(default 1000)		
3053	2	A-phase active power	-1200~1200 (-120%~120%) (default 50)	INT16	HEX
3054	2	B-phase active power	-1200~1200 (-120%~120%) (default 50)	INT16	HEX
3055	2	C-Phase active power	-1200~1200 (-120%~120%) (default 50)	INT16	HEX
3056	2	A-phase reactive power	-1200~1200 (-120%~120%) (default 0)	INT16	HEX
3057	2	B-phase reactive power	-1200~1200 (-120%~120%) (default 0)	INT16	HEX
3058	2	C-Phase reactive power	-1200~1200 (-120%~120%) (default 0)	INT16	HEX
3059	2	A-phase power factor	-1000~1000 (-1~1) (default 1000)	INT16	HEX
3060	2	B-phase power factor	-1000~1000 (-1~1) (default 1000)	INT16	HEX
3061	2	C-Phase power factor	-1000~1000 (-1~1) (default 1000)	INT16	HEX
3062	2	Active rate of change	1~30000 (1%/s~30000%/s) (default 2000)	UINT16	HEX
3063	2	Grid over-voltage level 1	105~150 (105%~150%) (default 110)	UINT16	HEX
3064	2	Grid over-voltage level 2	105~150 (105%~150%) (default 120)	UINT16	HEX
3065	2	Grid over-voltage class 3	105~150 (105%~150%) (default 130)	UINT16	HEX
3066	2	Grid over-voltage level 4	105~150 (105%~150%) (default 135)	UINT16	HEX
3067	2	Grid over-voltage level 5	105~150 (105%~150%) (default 140)	UINT16	HEX
3068	2	Grid over-voltage recovery	105~150 (105%~150%) (default 108)	UINT16	HEX
3069	2	Grid under-voltage level 1	10~95 (10%~95%)	UINT16	HEX

			(default 85)		
3070	2	Grid under-voltage class 2	10~95 (10%~95%) (default 80)	UINT16	HEX
3071	2	Grid under-voltage 3	10~95 (10%~95%) (default 60)	UINT16	HEX
3072	2	Grid under-voltage level 4	10~95 (10%~95%) (default 40)	UINT16	HEX
3073	2	Grid under-voltage level 5	10~95 (10%~95%) (default 20)	UINT16	HEX
3074	2	Grid under-voltage recovery	10~95 (10%~95%) (default 87)	UINT16	HEX
3075	2	Grid over-frequency level 1	20~500 (+0.2 ~ +5Hz) (default 100)	UINT16	HEX
3076	2	Grid over-frequency level 2	20~500 (+0.2 ~ +5Hz) (default 200)	UINT16	HEX
3077	2	Grid over-frequency III	20~500 (+0.2 ~ +5Hz) (default 300)	UINT16	HEX
3078	2	Grid over-frequency level 4	20~500 (+0.2 ~ +5Hz) (default 400)	UINT16	HEX
3079	2	Grid over-frequency level 5	20~500 (+0.2 ~ +5Hz) (default 500)	UINT16	HEX
3080	2	Grid over-frequency recovery	20~500 (+0.2 ~ +5Hz) (default 80)	UINT16	HEX
3081	2	Grid under-frequency level 1	-20~-500 (-0.2 ~ -5Hz) (default -100)	INT16	HEX
3082	2	Grid under-frequency II	-20~-500 (-0.2 ~ -5Hz) (default -200)	INT16	HEX
3083	2	Grid under-frequency III	-20~-500 (-0.2 ~ -5Hz) (default -300)	INT16	HEX
3084	2	Grid under-frequency level 4	-20~-500 (-0.2 ~ -5Hz) (default -400)	INT16	HEX
3085	2	Grid under-frequency level 5	-20~-500 (-0.2 ~ -5Hz) (default -500)	INT16	HEX
3086	2	Grid under-frequency	-20~-500 (-0.2 ~ -5Hz)	INT16	HEX

		<i>recovery</i>	<i>(default -80)</i>		
3087 3088	4	<i>Grid over-voltage protection time</i>	<i>0~3600000 (ms) (default 15000)</i>	<i>UINT32</i>	<i>HEX Hi-Lo</i>
3089 3090	4	<i>Grid over-voltage secondary protection time</i>	<i>0~3600000 (ms) (default 2000)</i>	<i>UINT32</i>	<i>HEX Hi-Lo</i>
3091 3092	4	<i>Grid over-voltage three-level protection time</i>	<i>0~3600000 (ms) (default 100)</i>	<i>UINT32</i>	<i>HEX Hi-Lo</i>
3093 3094	4	<i>Grid over-voltage four-level protection time</i>	<i>0~3600000 (ms) (default 100)</i>	<i>UINT32</i>	<i>HEX Hi-Lo</i>
3095 3096	4	<i>Grid over-voltage five-level protection time</i>	<i>0~3600000 (ms) (default 20)</i>	<i>UINT32</i>	<i>HEX Hi-Lo</i>
3097 3098	4	<i>Grid under-voltage protection time</i>	<i>0~3600000 (ms) (default 5000)</i>	<i>UINT32</i>	<i>HEX Hi-Lo</i>
3099 3100	4	<i>Grid under-voltage secondary protection time</i>	<i>0~3600000 (ms) (default 3000)</i>	<i>UINT32</i>	<i>HEX Hi-Lo</i>
3101 3102	4	<i>Grid under-voltage three-level protection time</i>	<i>0~3600000 (ms) (default 2000)</i>	<i>UINT32</i>	<i>HEX Hi-Lo</i>
3103 3104	4	<i>Grid under-voltage four-level protection time</i>	<i>0~3600000 (ms) (default 1500)</i>	<i>UINT32</i>	<i>HEX Hi-Lo</i>
3105 3106	4	<i>Grid under-voltage five-level protection time</i>	<i>0~3600000 (ms) (default 1000)</i>	<i>UINT32</i>	<i>HEX Hi-Lo</i>
3107 3108	4	<i>Grid over-frequency first level protection time</i>	<i>0~3600000 (ms) (default 30000)</i>	<i>UINT32</i>	<i>HEX Hi-Lo</i>
3109 3110	4	<i>Grid over-frequency secondary protection time</i>	<i>0~3600000 (ms) (default 10000)</i>	<i>UINT32</i>	<i>HEX Hi-Lo</i>
3111 3112	4	<i>Grid over-frequency three-level protection time</i>	<i>0~3600000 (ms) (default 5000)</i>	<i>UINT32</i>	<i>HEX Hi-Lo</i>
3113	4	<i>Grid over-frequency</i>	<i>0~3600000 (ms)</i>	<i>UINT32</i>	<i>HEX</i>

3114		<i>four-level protection time</i>	<i>(default 1000)</i>		<i>Hi-Lo</i>
3115	4	<i>Grid over-frequency</i>	<i>0~3600000 (ms)</i>	UINT32	HEX
3116		<i>five-level protection time</i>	<i>(default 100)</i>		Hi-Lo
3117	4	<i>Grid under-frequency first</i>	<i>0~3600000 (ms)</i>	UINT32	HEX
3118		<i>level protection time</i>	<i>(default 30000)</i>		Hi-Lo
3119	4	<i>Grid under-frequency</i>	<i>0~3600000 (ms)</i>	UINT32	HEX
3120		<i>secondary protection time</i>	<i>(default 10000)</i>		Hi-Lo
3121	4	<i>Grid under-frequency</i>	<i>0~3600000 (ms)</i>	UINT32	HEX
3122		<i>three-level protection time</i>	<i>(default 5000)</i>		Hi-Lo
3123	4	<i>Grid under-frequency</i>	<i>0~3600000 (ms)</i>	UINT32	HEX
3124		<i>four-level protection time</i>	<i>(default 1000)</i>		Hi-Lo
3125	4	<i>Grid under-frequency</i>	<i>0~3600000 (ms)</i>	UINT32	HEX
3126		<i>fifth level protection time</i>	<i>(default 100)</i>		Hi-Lo
3127	4	<i>Grid restoration</i>	<i>0~3600000 (ms)</i>	UINT32	HEX
3128		<i>confirmation time</i>	<i>(default 30000)</i>		Hi-Lo
3129	2	<i>Low penetration reactive current coefficient</i>	<i>0~600(0~6) (default 160)</i>	UINT16	HEX
3130	2	<i>Low penetration active current coefficient</i>	<i>0~100(0~1) (default 100)</i>	UINT16	HEX
3131	2	<i>Low penetration active recovery rate</i>	<i>1~30000(1~30000%/s) (default 100)</i>	UINT16	HEX
3132	2	<i>High penetration reactive current coefficient</i>	<i>0~600(0~6) (default 160)</i>	UINT16	HEX
3133	2	<i>High Throughput Active Coefficient</i>	<i>0~100(0~1) (default 100)</i>	UINT16	HEX
3134	2	<i>Battery Type</i>	<i>0: lithium, 1: lead acid, 2: PV, 3: DC bus (default 0)</i>	UINT16	HEX
3135	2	<i>Battery under-voltage protection point</i>	<i>2000~18000 (200~1800V) (default 6000)</i>	UINT16	HEX
3136	2	<i>Battery under-voltage recovery point</i>	<i>2000~18000 (200~1800V)</i>	UINT16	HEX

			(default 6200)		
3137	2	Battery over-voltage protection point	2000~18000 (200~1800V) (default 9000)	UINT16	HEX
3138	2	Battery over-voltage Recovery Point	2000~18000 (200~1800V) (default 8800)	UINT16	HEX
3139	2	Battery charge current limit	0~4000(0~400A) (default 2000)	UINT16	HEX
3140	2	Battery discharge current limit	0~4000(0~400A) (default 2000)	UINT16	HEX
3141	2	Battery float voltage	2000~18000 (200~1800V) (default 8700)	UINT16	HEX
3142	2	Battery equalization voltage	2000~18000 (200~1800V) (default 6300)	UINT16	HEX
3143	2	DC bus voltage reference	3000~18000 (300~1800V) (default 7500)	UINT16	HEX
3144	2	Insulation protection threshold	1~1000(k Ω) (default 100)	UINT16	HEX
3145	2	DC bus over-voltage point	3000~18000 (300~1800V) (default 9500)	UINT16	HEX
3146	2	Rated power	100~3000 (10~300kW) (default 500)	UINT16	HEX
3147	2	Rated phase voltage	100~600 (V) (default 230)	UINT16	HEX
3148	2	Rated frequency	50~60 (Hz) (default 50)	UINT16	HEX
3149	2	Grid-side wiring	0: 3-wire, 1: 4-wire (default 0)	UINT16	HEX
3150	2	Low penetration enable	0: disable, 1: enable (default 0)	UINT16	HEX

3151	2	High penetration enable	0: disable, 1: enable (default 0)	UINT16	HEX
3152	2	Reactive Mode	0: non-adjustable, 1: power factor, 2: reactive power (default 2)	UINT16	HEX
3153	2	Active anti-islanding	0: disable, 1: enable (default 0)	UINT16	HEX
3154	2	Insulation detection	0: disable, 1: enable (default 0)	UINT16	HEX
3155	2	Priority	0: Active priority, 1: reactive priority (default 0)	UINT16	HEX
3156	2	PV power low shutdown	0: disable, 1: enable (default 1)	UINT16	HEX
3157	2	On/Off Command	0: On-grid, 1: Off-grid (default 0)	UINT16	HEX
3158	2	Grid-connected switching	0: Command Trigger, 1: Grid Trigger (default 0)	UINT16	HEX
3159	2	Grid-connected control mode	0: PQ, 1: MPPT, 2: CV, 3: VSG (default 0)	UINT16	HEX
3160	2	Off-grid control mode	0: VSG, 1: VF (default 1)	UINT16	HEX
3161	2	Reserved	0~65535 (default 0)	UINT16	HEX
3162	2	Reserved	0~65535 (default 0)	UINT16	HEX
3163	2	Reserved	0~65535 (default 0)	UINT16	HEX
3164	2	Reserved	0~65535 (default 0)	UINT16	HEX
3165	2	Authorized Capacity	100~3000 (10~300kW) (default 500)	UINT16	HEX
3166	2	Inertia time constant	0~10000 (0~100s) (default 500)	UINT16	HEX
3167	2	Damping factor	0~30000 (0~3000) (default 500)	UINT16	HEX
3168	2	Primary FM enable	0: disable, 1: enable (default 0)	UINT16	HEX
3169	2	Primary voltage	0: disable, 1: enable	UINT16	HEX

		<i>regulation enable</i>	<i>(default 0)</i>		
3170	2	<i>Active FM coefficient</i>	<i>0~10000 (0~1000) (default 200)</i>	<i>UINT16</i>	<i>HEX</i>
3171	2	<i>FM frequency dead band</i>	<i>0~2000 (0~2Hz) (default 30)</i>	<i>UINT16</i>	<i>HEX</i>
3172	2	<i>FM active upper limit</i>	<i>0~2000 (0%~100%) (default 1000)</i>	<i>UINT16</i>	<i>HEX</i>
3173	2	<i>FM active lower limit</i>	<i>-2000~0 (-100%~0%) (default -1000)</i>	<i>INT16</i>	<i>HEX</i>
3174	2	<i>Reactive regulator coefficient</i>	<i>0~1000 (0~100) (default 125)</i>	<i>UINT16</i>	<i>HEX</i>
3175	2	<i>Voltage deadband</i>	<i>0~1000 (0~100V) (default 50)</i>	<i>UINT16</i>	<i>HEX</i>
3176	2	<i>Upper limit of regulating reactive power</i>	<i>0~2000 (0%~100%) (default 1000)</i>	<i>UINT16</i>	<i>HEX</i>
3177	2	<i>Regulating reactive power lower limit</i>	<i>-1000~0 (-100%~0%) (default -1000)</i>	<i>INT16</i>	<i>HEX</i>
3178	2	<i>Charging method</i>	<i>0: two-part, 1: three-part (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3179	2	<i>Reserved</i>	<i>0~65535 (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3180	2	<i>Reserved</i>	<i>0~65535 (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3181	2	<i>Reserved</i>	<i>0~65535 (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3182	2	<i>Module address⁽¹⁾</i>	<i>1~15 (default 1)</i>	<i>UINT16</i>	<i>HEX</i>
3183	2	<i>Cabinet address</i>	<i>1~10 (default 1)</i>	<i>UINT16</i>	<i>HEX</i>
3184	2	<i>Battery SOC</i>	<i>0~1000 (0%~100%) (default 0)</i>	<i>UINT16</i>	<i>HEX</i>
3185	2	<i>Battery SOC sag</i>	<i>0: Disabled, 1: Enabled (default 0)</i>	<i>UINT16</i>	<i>HEX</i>

Notes:

(1) The module address can only be adjusted in the non-parallel state using node number 160, and only IP65 modules support this feature.

4.5.2. Communication parameter setting

<i>ID</i>	<i>Byte</i>	<i>Item</i>	<i>Value</i>	<i>Value Type</i>	<i>Format</i>
3301	2	Dry contact 1 default	"0: Normally open" "1: Normally closed" (default 0)	INT16	HEX
3302	2	Dry contact 1 enable	"0: Disable" "1: Enable" (default 0)	INT16	HEX
3303	2	Dry contact 2 default	"0: Normally open" "1: Normally closed" (default 0)	INT16	HEX
3304	2	Dry contact 2 enable	"0: Disable" "1: Enable" (default 0)	INT16	HEX
3305	2	BMS enable	"0: Disable" "1: Enable" (default 0)	INT16	HEX
3306	2	BMS communication fault shutdown enable	"0: Disable" "1: Enable" (default 0)	INT16	HEX
3326	2	Communication Failure Confirmation Time	5~300 (5~300s) (default 30)	INT16	HEX

4.5.3. Example of Telecommand

Example: Modbus node number 01, write remote control information, ID is 3050, value is 100%, data order is H-L, checksum order is L-H.

Telecommand (HEX): 01 06 0B EA 03 E8 AA A4

Answer (HEX): 01 06 0B EA 03 E8 AA A4

4.6. Set date and time (remote control) — Function 03 or 16

<i>ID</i>	<i>Byte</i>	<i>Item</i>	<i>Value</i>	<i>Format</i>
1500	2	hour	0 : 23	HEX
1501	2	minute	0 : 59	HEX
1502	2	seconds	0 : 59	HEX
1503	2	year	0000: 2099	HEX

1504	2	month	1:12	HEX
1505	2	day	1:31	HEX

4.7. Remote On/Off (Remote Control) - Function 05 or 06

ID	Byte	Item	Value	Value Type	Format
5050	2	Switching on and off	0xFF00: Power on; 0x0000: power off	UINT16	HEX
5051	12	Standby	0xFF00: Standby; 0x0000: Shutdown	UINT16	HEX

4.7.1. Remote control example

Example: Modbus node number 01, remote control power-on command,
single word write without parameters, data order H-L,
checksum order L-H.

Remote control (HEX): 01 05 13 BA FF 00 A9 5B

Answer (HEX): 01 05 13 BA FF 00 A9 5B